ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B4 · RESOURCES

Renewable or finite?

★☆☆
Grade 4

Learning goals

- Sort resources.
- Name ways to conserve resources.

1 Sort the resources. [8]

SORT THESE

oil fish wind sunlight iron trees coal natural gas

Renewable (can be replaced)	Finite (will run out)
sunlight	natural gas
wind	coal
trees	oil
fish	iron

2 Match the action to the resource it saves. [4]

Draw a line to match each pair, or write the letter.

fixing a dripping tap ●	→ A	● **A** water
switching off the fan ●	→ C	● **B** fish
not catching young hilsa ●	→ B	● **C** electricity
using a cleaner cookstove ●	→ D	● **D** wood

3 Fill in the gaps. [3]

WORD BANK

wasting cycles finite conserving evaporating

Water is a resource that ___cycles___ through the environment. Gas and coal are ___finite___: they will run out. Using resources carefully so they last is called ___conserving___ them.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B4 · RESOURCES

Help or hindrance?

★★☆
Grade 4

Learning goals

- Evaluate technologies.
- Explain sustainable choices.

1 **Tube-wells gave millions of people in Bangladesh safe water, but some wells have arsenic. Is the tube-well a help or a hindrance? Explain both sides.** [3]

Help: it stopped many diseases from dirty pond water. Hindrance: some water has arsenic, which is a poison, and pumping lowers the groundwater. With testing and safe wells it is mostly a help.

2 **Why is it a problem if people catch fish faster than the fish can breed?** [2]

There will be fewer and fewer fish until they run out, so there is no food or work later.

3 **Natural gas will run out. Suggest two other ways Bangladesh could make electricity. Are they renewable?** [2]

Solar panels and wind turbines (renewable); also water power (renewable).

4 **Give an example of a sudden change and a gradual change to a resource.** [2]

Sudden: a cyclone destroys a forest. Gradual: groundwater gets lower each year.

ANSWER KEY

INVESTIGATE · B4 · RESOURCES

How much water does a dripping tap waste?

★★☆ Grade 4

Learning goals

- Measure volume and time.
- Use multiplication to scale up.
- Suggest action.

Class 4C put a measuring jug under a dripping tap for 10 minutes, three times.

1 **Complete the table.** [2]

Test	1	2	3	Mean
Water collected in 10 minutes (ml)	48	52	50	50

(48 + 52 + 50) ÷ 3 = 50 ml

2 **Use the mean to work out the water wasted.** [3]

Time	Water wasted
1 hour (6 × 10 minutes)	300 ml
1 day (24 hours)	7200 ml = 7.2 litres
1 week (7 days)	about 50 litres

3 **Why did they repeat the test three times?** [1]

To get more reliable results and find a mean.

4 **A family needs about 3 litres of drinking water each day for each person. How many people could drink the water wasted in one day?** [2]

7.2 ÷ 3 ≈ 2 people

5 **Write a message for your school about saving water.** [2]

e.g. "One dripping tap wastes 50 litres a week. Turn taps off tightly and report leaks!"

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B4 · RESOURCES

Catching the rain on the coast

★★★
Grade 4

Learning goals

- Understand a real water problem.
- Calculate rainwater from a roof.
- Evaluate a sustainable solution.

Science | Maths | Health | Design

In the south-west coast of Bangladesh, in places like Satkhira and Khulna, the water in ponds and under the ground has become **salty**. Sea water floods in during cyclones and moves into the ground. Drinking salty water is bad for health, so many women and girls walk long distances each day to fetch fresh water.

One solution is **rainwater harvesting**. Rain falls on a roof, runs into a gutter and pipe, and is stored in a big clean tank. The monsoon brings plenty of rain, so a family can store water to drink during the dry season. Rain is a **renewable** resource that cycles through the environment.

Based on information about water salinity and rainwater harvesting in coastal Bangladesh.

1 **Maths. 1 mm of rain falling on 1 square metre of roof gives 1 litre of water.** [3]

Roof area	Rain in one storm	Water collected
20 square metres	10 mm	200 litres
20 square metres	50 mm	1000 litres
40 square metres	50 mm	2000 litres

2 **Why is rainwater safer to drink than the pond water on the coast?** [2]

Rain is fresh water (not salty); stored in a clean, covered tank it stays free of salt and germs.

3 **How does a rainwater tank help girls and women?** [2]

They do not need to walk far to fetch water, so they have more time for school, work and rest.

4 **Design a rainwater harvesting system for your school. Label the roof, gutter, pipe, filter and tank.** [3]

Rainwater system

Roof → gutter → pipe → simple filter (cloth/sand) → covered tank with a tap; first flush thrown away.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B4 · RESOURCES

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

T	A	W	R	A	I	N	L	D	I
S	V	A	R	E	C	Y	C	L	E
G	L	R	E	S	O	U	R	C	E
A	J	F	I	S	H	W	E	R	J
S	J	F	M	T	R	A	F	B	H
F	O	R	E	S	T	T	K	N	C
R	B	E	E	A	D	E	G	L	H
M	C	O	N	S	E	R	V	E	T
C	V	H	O	V	K	R	V	G	B
Y	L	M	S	O	L	A	R	D	V

conserve fish forest gas rain recycle resource solar water

2 Which resource is finite? [1]

- ○ sunlight
- ○ wind
- ● **natural gas**
- ○ rain

3 Which is a sustainable way to fish? [1]

- ○ use nets with tiny holes
- ○ catch fish all year
- ● **stop fishing while fish lay eggs**
- ○ use poison

4 Draw the water cycle and label: evaporation, condensation, rain, rivers. [4]

Water cycle

Sun heats sea → evaporation → clouds (condensation) → rain → rivers flow back to the sea.

5 **Challenge. "Trees are renewable, so we can cut down as many as we like." Do you agree? Explain.** [2]

No. Trees are only renewable if we plant new ones and let them grow; if we cut them faster than they grow, forests disappear.